

School of Computer Science and Engineering, UNSW



WK07-01: Design Exercise

Software Architecture for Blockchain Applications,
COMP6452

Never Stand Still

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Design Exercise – Making Money Smart

For your project, aim to be able to do the following:

- Determine suitability of blockchains/DLTs to solve a constrained problem
- Select a blockchain platform to implement solution
- Draw a software architecture design to solve problem
- Evaluate design based on software qualities

Making Money Smart Project

NDIS (National Disability Insurance Scheme) – With greater choice & control, comes challenges

Challenge 1



Plan/Budget information isn't always automatically available in real-time

Challenge 2



Service eligibility determinations aren't always straightforward

Challenge 3



Payment interfaces & reconciliation can be complex for providers

Challenge 4



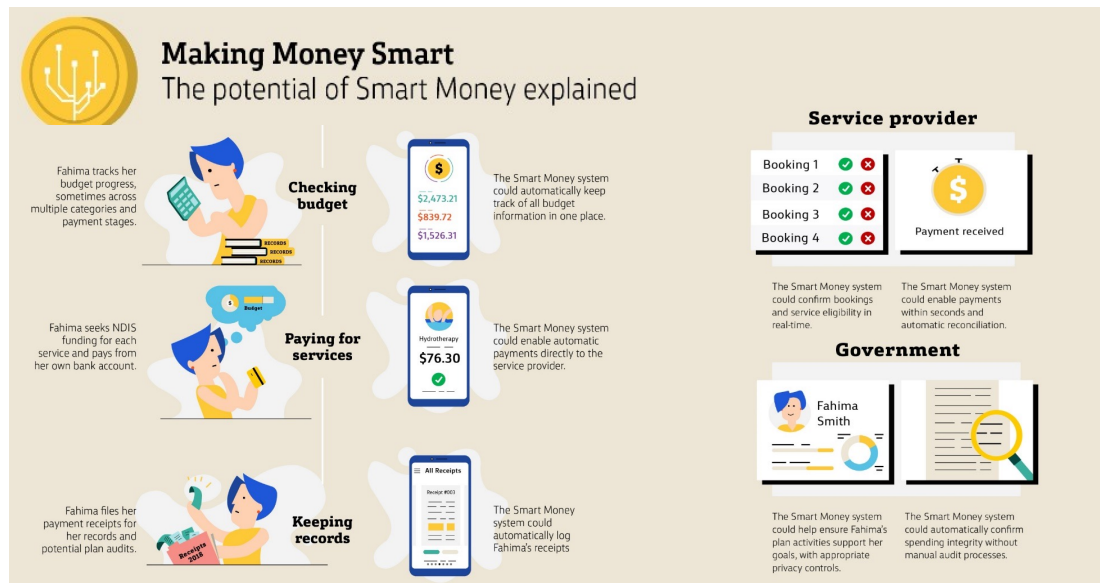
Misspending risks must be identified through manual audit activity

Challenge 5



Analytic's potential to improve participant outcomes isn't reached

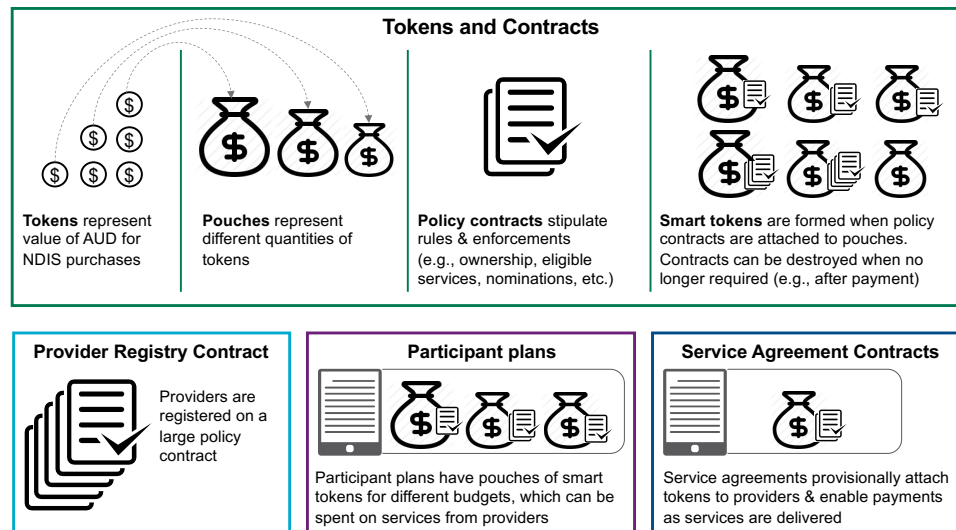
- NDIS (National Disability Insurance Scheme) provide greater choice and control to the participants. However, it comes with challenges to both participants & providers
 - What it could be spent on?
 - Who it could be spent by?
 - If they could be spent?
- July 2020 covers 460,000 Australians with greater choice & control over their disability supports. An annual investment of \$22 billion.
- See <https://data61.csiro.au/en/Our-Research/Our-Work/SmartMoney> for a detailed report



[See Making_money_smart_report.pdf](#)

- Smart money could achieve various benefits
- Participants can identify where and how much they can spend. They have a real-time view of how much of their allocation is left and simplify record keeping
- Providers can be sure that they are providing services for eligible participants reducing reimbursement risks. Also, simplify record keeping
- Government has better visibility on how the support money is spent and the compliance of both participants and providers

Making Money Smart – Tokens & Contracts



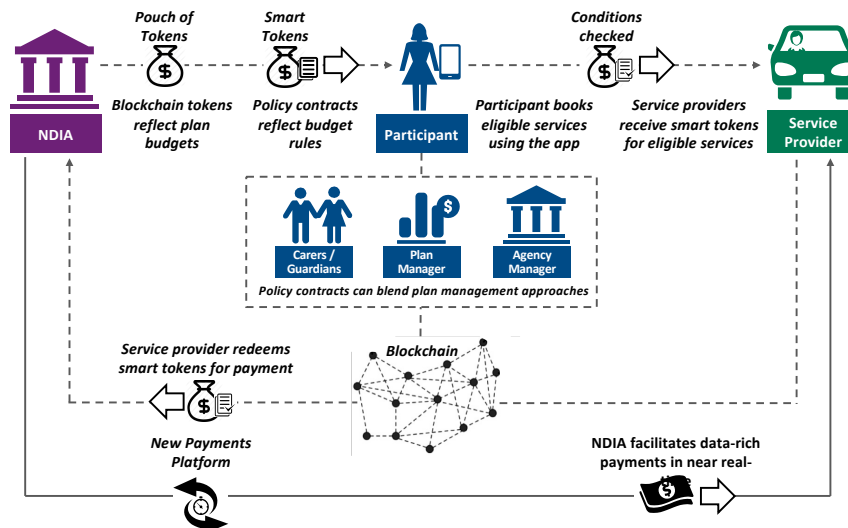
WK 07-01

Source: Daniel Royal et al., "Making Money Smart", 2018



- A token represent a certain no of AUD, e.g., 1 tokens – 1 AUD
- Multiple tokens are assigned to a pouch
- One of more policies are assigned to a pouch to denote ownership and payment conditions. Tokens in a pouch with attracted policy contracts are called smart tokens
- Provider information is maintained in a registry
- They would also get into a service agreement contract to get paid when conditions are satisfied
- Participant plans have pouches of smart tokens for different budgets (types of services), which can be spent on services from providers
- Each party had 3 budget lines and different addresses were assigned to each address

Making Money Smart PoC Architecture



WK 07-01

Source: Daniel Royal et al., "Making Money Smart", 2018

6



- NDIA - National Disability Insurance Agency
- NPP - Australia's New Payments Platform
- Participants never see the tokens - only their budget balances. Tokens are issues against those allowances
- At the time of booking a service budget rules are checked to see whether tokens can be spend on a certain provider for a specified service
- Once the service is provided, provider redeems token for AUD
- Australia's New Payments Platform to enable near real-time payments

Functional Requirements

1. The platform must onboard NDIS participants and service providers.
2. The platform must allow the NDIA to issue tokens to participants under different budget categories.
3. The platform must be able to add data about services (e.g., catalogues & service types).
4. The platform must allow NDIA to specify under what conditions those tokens can be spend by a participant.
5. The platform must allow NDIA to specify under what conditions those tokens can be accepted by a service provider.
6. The platform must allow NDIS participants to transfer tokens to eligible service providers.
7. The platform should allow a service provider to redeem tokens for fiat currency.

Non-Functional Requirements

Security – The platform must ensure that the identity of a stakeholder is proven to be the one claimed. Moreover, the platform must prevent unauthorised modification of data.

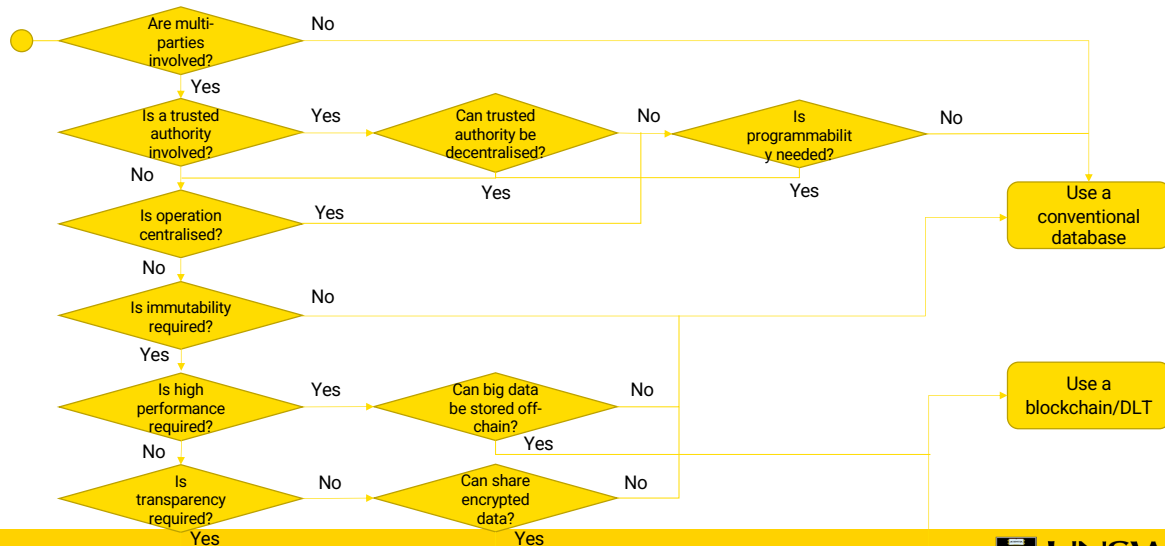
Privacy – Participant, medical data, & payment data must be visible only to NDIA and a selected service provider on a need-to-know basis (e.g., for a budget/service category).

Accountability – Token issue, transfer, and redeem must be immutable recorded.

Availability – System must be available 24x7.

Scalability – The system must support 500,000 users and 100 concurrent transactions.

Blockchain Suitability Assessment (One way)



Blockchain Platform Selection

Permissioned or permissionless?

Public or private?

- What level of privacy? Ledger state, TX data, all TXs?

Performance

Libraries & tool support

Prior experience & use cases

Platform stability – Is it time tested?

Integration with other enterprise systems & IT infrastructure

See S. Nanayakkara et al, "A methodology for selection of a Blockchain platform to develop an enterprise system", Journal of Industrial Information Integration, 23, 2021.

WK 07-01

10



- Permissionless - No authorisation required to perform a particular activity
- Permissioned - Authorisation required to perform a particular activity(ies)
- Public - Accessible to the public for use
- Private - Accessible only to a limited group of users for use

Some Design Questions to Ask

Who will feed data in & take out?

- Who should run a blockchain node/client?

What's on-chain & off-chain?

Do we need off-chain storage?

Do we need an oracle?

What smart contract?

- Tokens, business logic, registries, data only contracts, on-chain oracles, proxies, ...

Do we need a wallet?

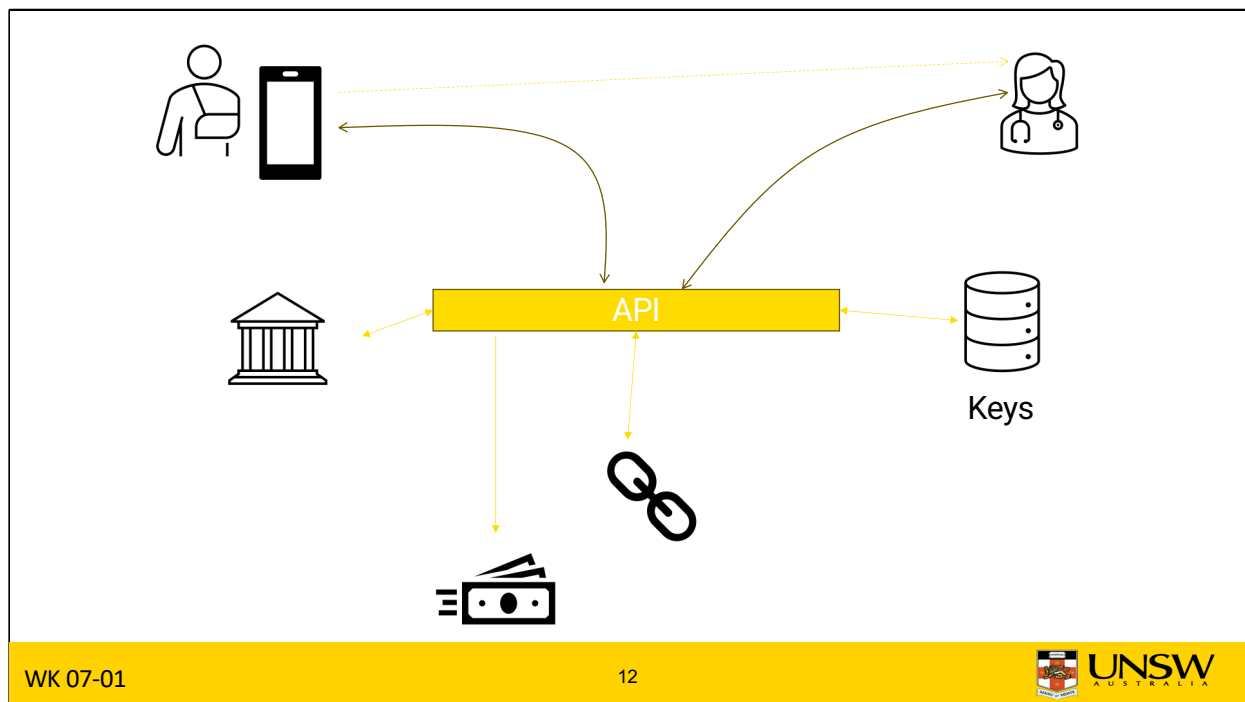
What other integrations are needed?

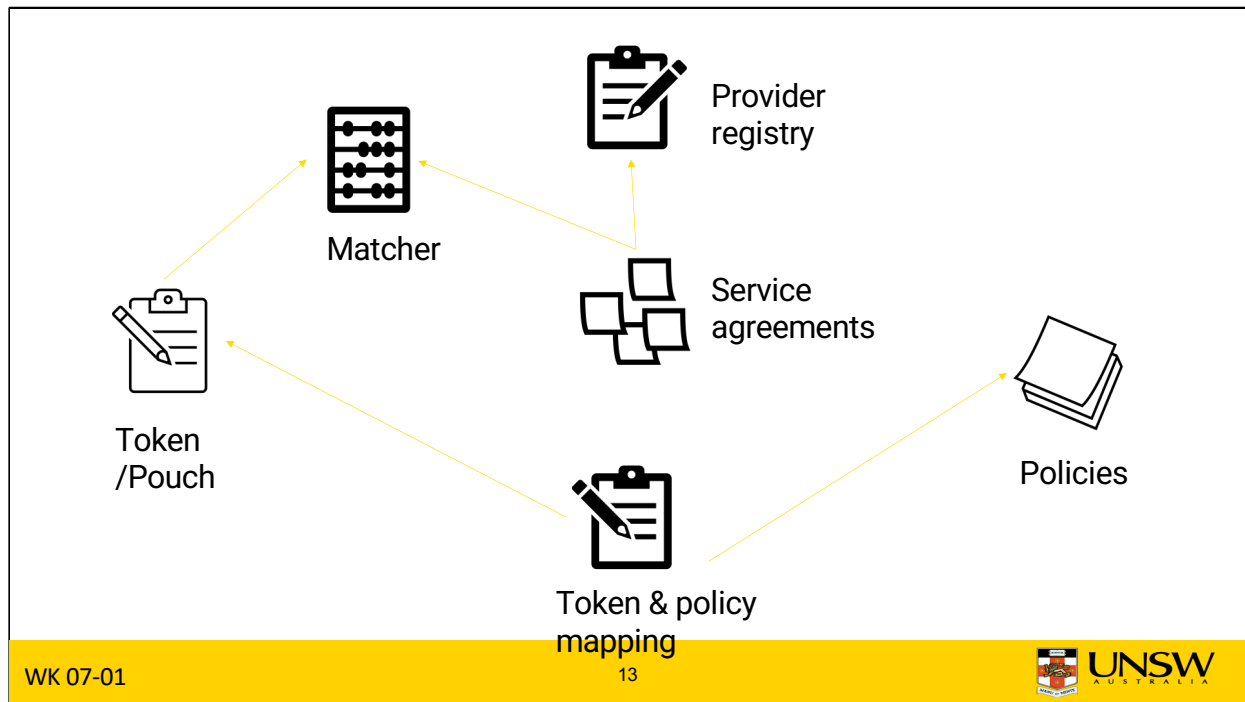
What data need to be private

Where to host?

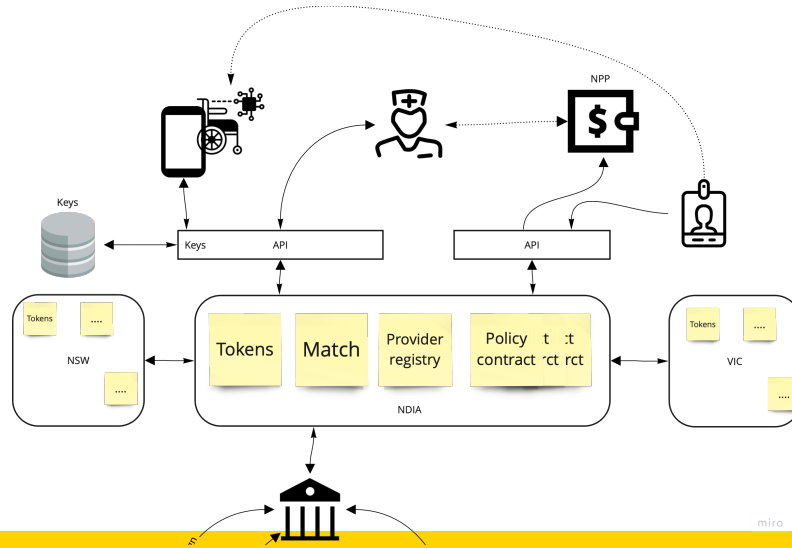
- Public, self-hosting, cloud hosting, blockchain as a service

- Not necessarily in a particular order

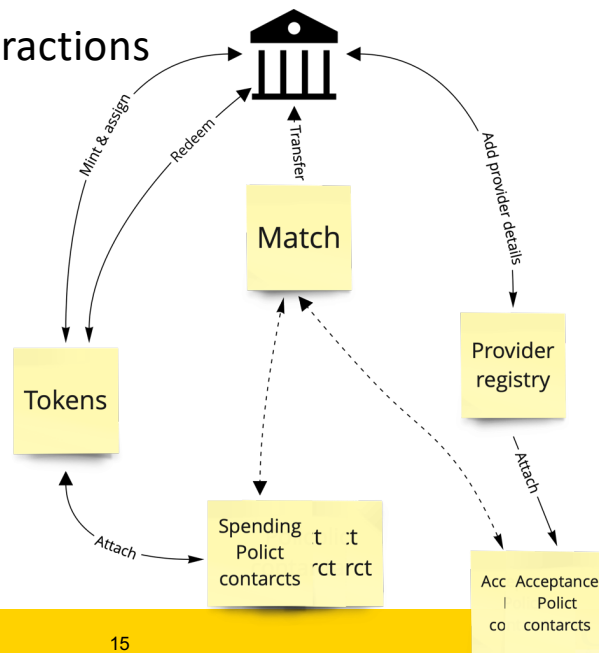




A Possible Design



Smart Contract Interactions



If Parties Had Similar Authority?

